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Interactional metadiscourse in research article abstracts

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Abstract

This paper deals with interpersonality in research article abstracts analysed in terms of interactional metadiscourse. The evolution in the distribution of three prominent interactional markers comprised in Hyland's (2005a) model, viz. hedges, boosters and attitude markers, is investigated in three decades of abstract writing in the field of applied linguistics in the broad sense. On the basis of a quantitative corpus survey of abstracts in *Journal of Pragmatics*, two major points are made. One is that the distribution of hedges, boosters and attitude markers in abstracts, when compared with their distribution in research articles, supports the idea that abstracts are not just pale reflections of the full-length articles, but rather have a specific make-up, which can plausibly be linked to their function. The second point is that the use of interactional metadiscourse in abstracts has undergone interesting changes in the course of the past 30 years. On the whole, the degree of interpersonality realised by hedges, boosters and attitude markers diminishes over time, though notable differences exist with regard to the subcategories in the interactional domain. In the discussion section, we try to arrive at an explanation for the changes that have occurred, taking genre, discourse community, research practice and rhetoric strategy into account.
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1. Introduction

1.1. The RA abstract as a genre

Research article abstracts (RA abstracts) have become a well-established genre in academic discourse since Eija Ventola's (1994) plea for a linguistic approach to the genre that was capable of combining a global structural view of the genre with a prescriptive local view of the linguistic realisation of the abstract. The question of whether a RA abstract functions as a condensed reproduction of the text or rather as an expansion of the title, as well as the question of whether it is an indicator of the RA's content or rather an informative summary remain a matter of ongoing debate (Ayers, 2008; Hyland, 2000; Lorés, 2004; Stotesbury, 2003; Swales, 1990; Ventola, 1994; Yakhontova, 2002). There is, however a broad consensus on the abstract's status as a separate genre. Indeed, since the late seventies RA abstracts have become part of a standard policy in article publication, and the discourse community of scholars, confronted with

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an immense amount of output in a globalised publication market, has asked for RA abstracts as an orienting tool for their reading (Salager-Meyer 1990, p. 366; Ventola 1994, p. 333; Lorés 2004, p. 281). As Miller (1984) in her pioneering and often cited article on 'Genre as social action' demonstrated, genres emerge from the requirements of recurrent rhetorical situations, asking for an adequate response. The RA abstract is such a response, and, by nature, a social one. From this viewpoint the generic character of the RA abstract is no longer defined in terms of specific lexis and syntax, but rather as a case of interaction between individuals, acting in a social, institutional context. In this light the interpersonal dimension of the RA abstract becomes evident, and this is why Hyland (2005a) highlights the interpersonal character of all metadiscoursal devices in academic texts, including RA abstracts. The approach followed in this article links up with this research, as it concentrates on the linguistic markers of interpersonality in RA abstracts, whereby interpersonality is explored by using Hyland's (2005a) categorisation of interactional metadiscourse as an operational model.

This particular perspective on interpersonality is complementary to earlier research, which has mainly focused on the study of the move structure of particular genres, especially since Swales's 1990 seminal publication on genre. The latter line of research was needed because the IMRD structure of the abstract proved to be less evident than was first assumed (Graetz, 1985; Salager-Meyer, 1990) and in fact its move structure displayed much variation, as shown by Hyland (2000) and Samraj (2005). Dahl (2004) and Lorés (2004) as well have recently amended the classical four-part rhetorical structure of RA abstracts. Moreover, based on their research findings the abstract appeared to be a distinct genre, not merely replicating the research article, but enhancing its news value by focusing on the results and downgrading or even omitting the method move. Lorés quite rightly pointed at the problem of subjectivity in the move identification and suggested another structure, the so-called CARS structure, based on Swales's move analysis of the introductions of research articles.

In contrast to its move structure, the language of the RA abstract has received little attention. According to the ANSI (American National Standards Institute) guidelines the RA abstract should be written in an objective mode, and Graetz (1985, p. 125) points out that the abstract is characterized by 'tightly worded sentences' which avoid 'repetition, meaningless expressions, superlatives, adjectives (…)', and so on and by, for example, the predominance of nominalisations and passive voice, all of which can be considered as standard characteristics of academic prose in general. Research from Bazerman (1984) to Hyland (2005a) and Biber (2006) has shown, however, that upon closer scrutiny academic writing is deeply intertwined with subjectivity. The expression of evaluation, of stance, and engagement has become evident in academic genres, not only in RAs, but in RA abstracts as well (see e.g. Stotesbury, 2003).

Another point of concern is the historical, dynamic nature of genres. Conventionalised wisdom seems to maintain that RA abstracts form a stable and fixed genre. However, just like other genres, they might show variation and evolution. There has not been much research into the evolutionary nature of abstracts. Ayers (2008), when dealing with this issue, focuses on structure again, not on language. As Hyland (2005a, p. 202) rightly states: "(…) there are very few diachronic studies of metadiscourse practices and research is urgently needed to document changing thought styles, patterns of arguments, and ideological practices over time."

In this paper we aim to investigate (1) to what extent RA abstracts differ from RAs with regard to the use of interpersonal elements, more specifically with the use of prominent interactional features as defined by Hyland (2005a), and (2) whether any changes have occurred in the past 30 years, since the use of RA abstracts has become customary. In this way we hope to shed new light on the crystallisation of RA abstracts as a separate, full-fledged genre in academic discourse, separate from the RA and more specifically from the RA's introduction, out of which it has evolved.

1.2. Interpersonality in RA abstracts

Subjectivity in academic discourse has been studied from terminologically and conceptually different perspectives like appraisal, attitude, evaluation, stance, interpersonality etc. In this article, we take Hyland's interpersonal model of metadiscourse as a point of departure. According to Hyland and Tse (2004) and Hyland (2005a, p. 41) "**all** metadiscourse is interpersonal in that it takes account of the reader's knowledge, textual experiences and processing needs and that it provides writers with an armoury of rhetorical appeals to achieve this".

Still, following Thompson and Thetela (1995) and Thompson (2001), Hyland and Tse (2004) make a useful distinction between interactive and interactional metadiscourse. Both have an interpersonal function, but the former is concerned with guiding the reader through the text with the aid of signposts like transition markers, sequencers, code

glosses, frame markers etc. These interactive resources are comparable to what Halliday (1994) has called the textual metafunction. Interactional metadiscourse, on the other hand, deals with the expression of the opinion of the writers, and their relationship and interaction with their readers. Therefore interactional resources are more related to Halliday's interpersonal metafunction (Hyland, 2005a, p. 44). Whereas interactive metadiscourse "primarily involves the management of information flow" (Hyland, 2005a, p. 44), interactional metadiscourse is "more personal" (Hyland, 2005a, p. 44) and involves the reader more overtly in the text by commenting on and evaluating the text material. Interactive metadiscourse markers like connectives are more closely related to the discourse, and only in argumentative texts do they seem to fulfil an interpersonal function. Thus, as interactional metadiscourse is more directly and evidently related to interpersonality, we have restricted our research to this type.

As mentioned earlier, RA abstracts have been shown not to be fully objective summaries of the RAs they accompany, but rather to be replete with subjective material, expressed by interactional elements (Hyland, 2000, pp. 63–84). Not much is known, however, about the way they have evolved. It is not clear whether the practice of writing RA abstracts has changed from 1977 to 2007, and if so, whether writers have begun to use more interactional features over time, or rather have been using them more sparingly. As will be shown in the remainder of this paper, a diachronic approach indeed leads to a better understanding of the way RA abstracts are composed, how they function as a genre and how they constitute a rhetorical platform for interpersonal relationships.

2. Methodology

2.1. Quantitative research

In order to investigate how RA abstract writers deploy interactional metadiscourse, we have carried out a quantitative study. In this way the results can be compared to the findings on RAs that Hyland reports on in a number of studies (Hyland, 1998, 2005a, 2005b), although he stresses that his research on metadiscourse in academic writing is primarily qualitative in nature, and that the quantitative data he supplies have, above all, an illustrative function.

In our opinion a quantitative approach is indispensable if one is interested in the microvariation that is the focus of the present study: due to the succinct form of RA abstracts, familiarity with the genre on the part of the linguistic discourse community, and the explicit norms issued by the editorial boards, RA abstracts can be expected not to display much variation. RA abstracts are a highly conventionalised genre (Graetz 1985, p. 125) with a predominance of set expressions (e.g. "in this article it is argued …"). Moreover, the quarter of a century that RA abstracts have been around in the field of (applied) linguistics is so short a period that changes in the use of metadiscourse could be hard to discern with the naked eye. A quantitative study may be able to unveil small yet significant divergences of the standard patterns, which may be the result of conscious or unconscious decisions of RA abstract writers.

2.2. Corpus

The corpus we used consists of 72 RA abstracts from articles in *Journal of Pragmatics* taken from different volumes with a 5 year interval between each, starting in 2007 and going back to 1982, yielding a total of 11,141 words. Articles without an abstract, like book reviews and introductions to special issues, were ignored. For the selection of the articles, we followed a simple algorithm, selecting the nth abstract from issue n (so first abstract of issue 1, second abstract of issue 2 etc.). In cases where there was no such n^{th} abstract, that is when the issue number exceeded the number of articles in it, we looped over the available articles, so that the 8th article in an issue containing 5 articles-with-abstracts, would result in the actual selection of the 3rd article. In cases where the issue in question did not feature any articles with abstract, recourse was taken to the previous issue.

We made an additional sample extraction for the seventies, but as the first volume of the journal only appeared in 1977, we were not able to continue the five year interval procedure. As it was thus not possible to select abstracts from 1972, we selected twice 12 abstracts from 1977 and 1979 respectively. Due to this sampling problem, the data from the seventies will be dealt with in a separate section.

The reason for choosing *Journal of Pragmatics* is its high output rate (currently 12 issues per volume), its high prestige in the field, and the large discourse community it functions in, with articles from a wide range of scholars and subfields. Moreover, it is one of the very few journals that already featured RA abstracts in the early eighties. Other journals in the domain of applied linguistics (in the broad sense) lack RA abstracts in the early eighties (e.g. *Applied*

Linguistics), or were only recently launched (e.g. *Discourse Studies*). Also, by restricting ourselves to one particular journal we were able to control systematic variation due to editorial guidelines (stylistic and other) that the RA abstract authors are asked to comply with. On the negative side, extrapolations of our results to the whole field should be approached with more than due caution.

2.3. Detecting interactional metadiscourse

As previously said, we have stayed close to Hyland's (2005a) classification of interactional metadiscourse. He distinguishes five different types of interactional elements: attitude markers, boosters, engagement markers, hedges and self mentions.

Attitude markers express the opinion of the author. Boosters are used to strengthen a claim, or to close off alternative voices. Engagement markers stress the relationship between the author and the reader, e.g. by the use of inclusive *we*. Hedges let alternative voices come into play by putting the propositions under discussion. Self mentions are places where authors put themselves explicitly on stage. Respective corpus examples are given in (1)–(5).[1]

(1) ATTITUDE MARKER
This claim is **misleading**. (JoP 1982–2)
(2) BOOSTER:
[T]he distribution and choice of pronominal forms within these speeches **clearly** reflects the personal and political ideologies of the speakers. (JoP 1987–8)
(3) ENGAGEMENT MARKER:
To remedy this, **we** should view generations preeminently as a planning task. (JoP 1987–11)
(4) HEDGE:
All this **seems** to indicate that zero anaphora in Chinese, being realized by syntactically undifferentiated gaps rather than by specific empty syntactic categories, are not grammatically by pragmatically determined. (JoP 1992–1)
(5) SELF MENTION:
We give an historical overview which should explain why AI investigations in language developed apart from theoretical linguistics (JoP 1982–5)

Note that the pronoun *we* is used here to exemplify both the category of the engagement markers and the category of the self mentions. The difference is that we have an inclusive *we* in the former case, and an exclusive *we* in the latter.

In this paper, we have focused on three kinds of interactional elements, viz. hedges (H), boosters (B) and attitude markers (A). Self mentions have been ignored because there is no agreement on their interpersonal effect. According to Langacker (1990) for example, the explicit mention of first person pronouns leads to a *less* subjective, less interpersonal construction than when they are left implicit. Moreover, their use is highly conventionalised. First person pronouns are often shunned in favour of impersonal constructions, e.g. *it is argued*, rather than *I argue*. This genre convention does not necessarily reflect the interpersonal character of the RA abstract. Still, for certainty's sake, we have checked whether including self mentions in the counts would affect the results. This was not the case. The overall tendencies to be dealt with in Section 3 remain as they are.

Engagement markers were left out of the analysis because they are virtually absent anyway in RA abstracts, and because the few elements that may qualify as engagement markers are hardly distinguishable from attitude markers, as is also recognised by Hyland (2005a, p. 54). Again, to make sure, it was checked whether the quantitative findings would be affected if we had included the engagement markers. This turned out not to be the case.

In sum, hedges, boosters and attitude markers are in our view the most relevant interactional markers contributing to the interpersonal dimension of the RA abstracts. It is in these three subcategories that authors can deliberately choose how to dialogue with their readers by modifying their claims, through strengthening, downplaying and commenting.

For the detection of hedges, boosters and attitude markers the whole text was scrutinised word by word, rather than following the procedure by which one selects a number of typical hedges, boosters, and attitude markers from a list to

[1] The source of the examples is identified in the tag that follows it. *JoP* stands for *Journal of Pragmatics*. Then follows the year of the volume (1977–2007), and the abstract we sampled (1–12).

subsequently track them automatically in the corpus. In this way, we were able to include fewer typical interactional items as well.

(6) (…) proposals which, **in one way or another**, call for disagreement (…) (JoP 2002–11)

For instance, in (6), the prepositional phrase "in one way or another" functions as a hedge. In a list-based approach, starting from single lexemes associated with hedging (*perhaps*, *possibly*, *may* etc.), this particular example could have passed unnoticed.

The word-by-word procedure is not entirely without problems, though. One of the difficulties in the evaluation of the texts is the precise demarcation of the interactional units. This can be illustrated by the following example.

(7) **I would like to suggest** that counterfactual thinking is an essential necessity of human existence. (JoP 1997–12)

It is clear that this is a hedged version of the assertion that "counterfactual thinking is an essential necessity of human existence." But where is the hedge located exactly? Is it one hedge, idiomatically expressed as "I would like to suggest"? Or is it two hedges "would (like to)" and "suggest"? In these cases, we have decided to count only one hedge, and this was done consistently throughout the corpus.

3. Results

3.1. Synchronic analysis: a brief comparison of RA abstracts and RAs

In Figure 1 below, Hyland's (2005b, p. 187) results of his study on hedges, boosters and attitude markers in RAs in applied linguistics are contrasted with our counts. Although the comparison may be flawed by the fact that the assessment is not carried out by the same person and by the fact that the two corpora may not be entirely comparable,[2] there is a telling difference: whereas the total use of HBAs[3] (last column) is more or less comparable in both genres, as is the number of attitude markers, abstracts make far more use of boosters and far less use of hedges than research articles.

This does not come entirely as a surprise, in view of the functional difference between the two genres. Abstracts are intended to convince the reader that the article is worth reading and that the author presents interesting research findings. As a result, claims are likely to be emphasized by means of boosters, rather than downplayed by hedges. In general, hedges on scientific claims are likely to disappear when the claims are indirectly referred to, in other (academic and popularising) articles (De Waard, 2009). The same effect may occur in the transfer of the claims from to RAs to the RA abstracts.

3.2. Diachronic analysis

3.2.1. Length of the abstracts

Before going into details about the changes in the use of hedges, boosters and attitude markers, let us first take a look at the changes in the size of the RA abstracts. This can be of relevance for the present study as a change in the average length can be seen as a first indication that the genre of the RA abstract is somehow under pressure. As Graetz (1985, p. 125) points out, one of the hallmarks of the RA abstract as a genre is its succinct form and the consequential avoidance of redundancy. Moreover, it is not unusual for editorial boards to stipulate the maximum length. These constraints on the size of RA abstracts make it thus unlikely that their length varies randomly. As is clear from Figure 2 there is, however, a clear tendency in our corpus: the mean length of the RA abstracts tends to get longer over time, concurring with Hyland's (2000, p. 81) findings.

Apart from the mean length of the abstracts, Figure 2 also plots the standard deviations per decade. This tells us something about the spread among the abstracts: the larger the standard deviation, the less uniform the length of

[2] To improve the mutual comparability, we did not look at the frequencies in the whole JoP corpus, but only at those in the 90s sample.

[3] The abbreviation *HBA* (for hedges, boosters and attitude markers) in Figure 1 will be used throughout this article. *RA abstract* is abbreviated to *RAA* in the figures. *HBA per Kiloword* refers to the occurrences per 1,000 words of running text.

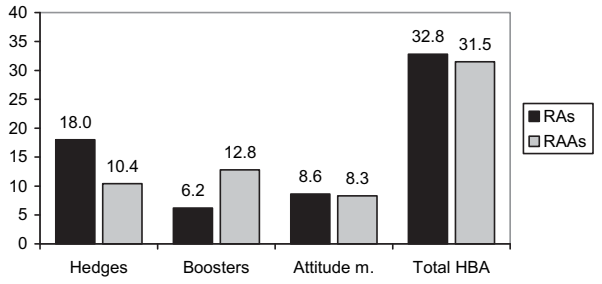


Fig. 1. Interactional markers per kiloword in RAs (Hyland) and RAAs (Gillaerts & Van de Velde).

the abstracts in that decade. Figure 2 shows that there is no straightforward tendency to be discerned in the data. In fact, the standard deviations remain fairly stable, suggesting that the rise in the mean length is a rather robust change.

The question now is what triggers the increase in length. Do RA abstract writers make more use of metadiscourse, or is there simply more factual material in the abstracts? We will come back to this issue in Section 4.

3.2.2. The use of interactional metadiscourse

Of more direct concern to our present research goal is the distribution of HBA-markers. A number of tendencies emerge from our results. Figure 3 shows that the overall use of the three interactional markers under investigation decreases appreciably over time.

Underneath this general tendency clear differences can be detected. The global decline of HBA-markers cannot be broken down in a series of fully uniform declines in the different categories. As can be seen in Figure 3, there is a sharp division between the hedges on the one hand, and the boosters and the attitude markers on the other. The continuous decrease of interactional metadiscourse as a whole in the abstracts is entirely due to the latter categories (boosters and attitude markers): whereas boosters and attitude markers are noticeably on the wane, the use of hedges does not seem to drop at all. Rather, their occurrence seems to rise throughout the various decades, albeit not in a linear way. Even when the top and bottom values are cut out (that is: both the abstract with the highest and lowest number of HBA-markers per kiloword, respectively), the analysis yields similar result, except for the hedges, which now indeed show a uniform rise over the three decades.

Notable differences between the use of hedges, boosters, and attitude markers also emerge in Figure 4, representing the *relative* use of these markers in each decade, making abstraction of the overall decline of the HBA-category.[4] It can be observed that the relative use of hedges rises noticeably, with a concomitant drop of the attitude marker category. Boosters, constituting the most popular class of interactional devices in any period as was already clear from Figures 1 and 3, neither gain nor lose relative importance over time. So, although in absolute terms boosters are cut back, their use in relation to the other interactional devices is not diminishing at all.

Overall, it seems that HBAs are more sparsely used in the recent abstracts in our corpus, and if they are used, there is a notable increase in the use of hedges.

3.2.3. The combination of hedges, boosters and attitude markers

The scholarly literature on metadiscourse mainly focuses on the use of hedges, boosters and attitude markers whose function is to alleviate, reinforce, or comment on propositional material. What is easily overlooked is that these interactional resources can also be used to modify each other. So instead of subduing a claim, a hedge can also be used to mitigate a booster (e.g. *a rather strong correlation*) or an attitude marker (e.g. *rather important findings*). In (8)-(9) we give some corpus examples.

[4] For the relative use of HBA-markers, the frequencies did not require to be normalised to 1,000 word samples, so that a straightforward statistical technique is available for testing dependency across strata, viz. the Cochran-Mantel-Haenszel 2 statistic testing the hypothesis that row mean scores differ. We also give the results for the well-known chi-square test.

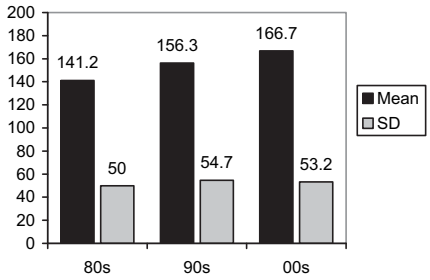


Fig. 2. Mean length (number of words) and standard deviation of the sampled RA abstracts.

(8) HEDGED ATTITUDE MARKER:
(…) that there **may (H)** be **virtue (A)** in the empirical, the particular and the factical (…) (JoP 1997–1)
(9) HEDGED BOOSTER:
(…) **it may be that (H)** the **biggest (B)** hurdle in developing computer programs which can successfully carry out conversations will be modeling the ability to 'cooperate'. (JoP 2002–3)

Table 1 gives an overview of the interactions of boosters and attitude markers with hedges in our corpus. First, the absolute frequencies of hedged boosters and unhedged boosters are given. The next two columns present the frequencies of hedged attitude markers, and attitude markers in isolation, respectively. The last two columns give the combined figures.

The trend becomes clearer when the absolute frequencies in table 1 are given as percentages, as shown in Figure 5. The results clearly show an increased popularity over time in the combined use of hedges, boosters and attitude markers. In other words: boosters and attitude markers are increasingly less often used on their own.

3.3. Metadiscourse in the seventies

As noted above, two additional samples were taken from the early volumes (1977-1979) of *Journal of Pragmatics*. The results of the analysis of these samples suggest that the tendencies that have been discussed above need to be modified in a number of ways.

As regards the mean length of the abstracts, a 140.6 score for the 70s sample fits neatly in the tendency mentioned above, that is: increasing length over time. The standard deviation of 65.5 on the other hand, is markedly higher than what was established for the 1982-2007 period. So, on average, abstracts are shorter in the early volumes, but there is considerable variation in the length of individual abstracts. An extreme example is the following abstract from 1979, consisting of no more than a single sentence.

(10) This paper lops off a reputed branch of the meaning of meaning by showing that meaning never means referring. (JoP 1979–10)

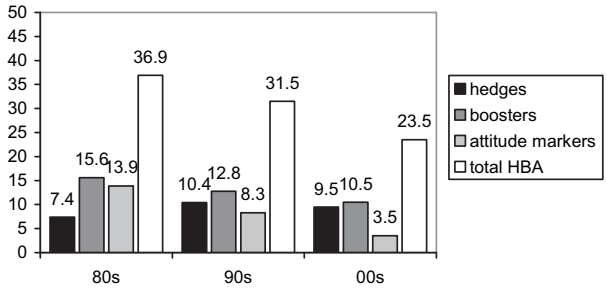


Fig. 3. Interactional metadiscourse per kiloword in RA abstracts from a diachronic perspective.

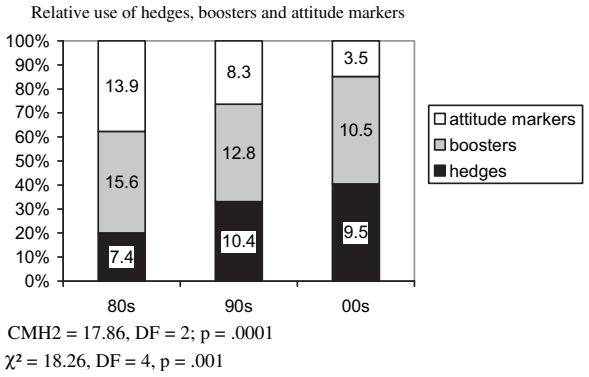


Fig. 4. Relative use of hedges, boosters and attitude markers.

A more complicated picture is obtained when the use of the three interactional resources is investigated in the additional sample. If the consistent tendency we observed in the later volumes extends to the first volumes as well, a rather abundant use of HBA-markers would be expected, but this is not always the case. For the use of hedges, the results (7.2 instances per kiloword) fit in with what is to be expected, but both boosters (9.1 instances per kiloword) and attitude markers (7.5 instances per kiloword) are visibly underrepresented. The inconsistency in the extra sample extends to the relative use of HBA-markers. The results (30.4% hedges (n = 24); 39.2% boosters (n = 31); 30.4% attitude markers (n = 24)) indicate that the late 70s score in between the 80s and the 90s sample, with a higher than expected proportion of hedges.[5]

It appears that the two 70s volumes are a bit 'ahead of their time' when it comes to the use of interactional markers. If we take into account the high standard deviation on the length as well, it looks as if authors allowed themselves more freedom in the way they devised an abstract. The rather laconic RA abstract in (10) is presumably less likely to pass the present-day review process. It seems that it took a few years before the RA abstracts became standardised.

4. Discussion

Two aims were set for the present paper. One was to provide some support for the idea that the abstract is a separate genre from its associated article. This idea can indeed be sustained by the distribution of the three interactional metadiscourse categories under investigation as interpersonality markers. RAs and RA abstracts show differential use of various subcategories of interactional metadiscourse: whereas research articles exhibit a rather high number of hedges in comparison to boosters and attitude markers, abstracts show more affinity with boosting, rather than with hedging. As was pointed out, this links up with the functional differences of both text types. In this respect, it is important to realise that summarising the RA is not the only function of RA abstracts. The latter also have a promotional function (see Yakhontova, 2002 for the difference between 'telling' and 'selling', and Dahl, 2004, pp. 59–60). Indeed, as Hyland (2000, pp. 64–65) indicates, the competitive nature of the research community causes RA abstracts to function as an advertising means for attracting readers to the full-length text of the research article, and this ties in with the observation that boosters are the most important means of interaction marking in RA abstracts in any period (see figure 1, 3 and 4).

Our results may be further refined by future research focusing on inter-generic relations between RAs and RA abstracts, in which the move analysis is also taken into account. Arguably, the interactional markers are not equally distributed over the various IMRD sections. Hedges probably occur more frequently in discussion sections than in methodology sections, as the discussion sections are more likely to contain controversial claims, which need more negotiation between writer and reader and accommodation of alternative voices. The extent to which the RA abstracts reflect the move structure of the RAs can impact the frequency and interpretation of the interactional markers.

[5] Still, statistical tests suggest that the dependency between the decade and the use of metadiscourse elements remains significant at a 1% alpha level. CMH2 = 10.20, DF = 2; $p = 0.006$ and $\chi^2 = 18.42$, DF = 6, $p = 0.005$.

Table 1
Interaction between hedges and boosters/attitude markers (absolute figures).

	B with H	B without H	A with H	A without H	AB with H	AB without H
80s	4	49	4	43	8	92
90s	8	40	10	21	18	61
00s	9	33	5	10	14	43

Still, there are some indications that the generic differences between RAs and RA abstracts have become smaller in recent years. As RA abstracts gain importance and sometimes even become the prime locus of knowledge exchange in the scientific discourse community, they function as 'mini-articles' to some extent. This is both supported by the increase in the average length of the RA abstracts and the decrease in interactional features. The longer RA abstracts in recent years do not display a higher density of interactional metadiscourse; nor is there a significant increase in interactive metadiscourse (see Gillaerts & Van de Velde, 2009, pp. 89–90, for detailed frequencies).

This means that the longer, recent RA abstracts contain more 'factual' material, instead of more metadiscourse. This holds true even if we take into account the omitted interactionals (self mentions and engagement markers). This increase of factual discourse material could be the result, we think, of the generic shift of RA abstracts. From an embedded genre (to use a term from Bhatia, 2004), functionally and structurally placed in between the title and (the introduction of) the RA and as a rule to be read in combination with the full-text article, they are gradually changing into a stand-alone genre, often consulted on their own by scholars who want to assess the RA's relevance for the field. Although this is not always considered good scientific practice, RA abstracts in many fields, including (applied) linguistics, are often consulted by practitioners in this way. As a result of the increase of scientific output, there is a rising need for quick-to-consult summaries of research results.

Moreover, RA abstracts are published in abstract databases or abstracting journals, which often make them freely available, as opposed to the full-text article. Such databases, in particular when accessed electronically, have been used to an increasing extent over the past decades.[6]

The stand-alone nature of the RA abstract is furthermore strengthened by its use in selection procedures. Acceptance of a paper for a conference crucially depends on the quality of the submitted abstract, and the same may be true for acceptance for publication in a journal. The summary function of the RA abstract can then get obscured, so that it becomes a 'promise', rather than a 'report'. This is formally reflected to some extent by the use of future tense verbs, as in the following examples:

(11) A major part of this paper is devoted to a presentation of our ideas about artificial intelligence. We **will** describe why we believe AI can make genuine and substantial contributions to the study of language (JoP 1982–5)
(12) This article presents arguments against Chomsky's (1982) quadripartite typology of empty categories. First, I **shall** argue that neither PRO nor pro can occur in Chinese. (JoP 1992–1)
(13) The analysis **will** show the difference in the use of these elements between both groups of speakers, with the possible pragmatic failures that may arise in communication (JoP 2002–6)

The second aim was to look for diachronic changes in the use of hedges, boosters and attitude markers in RA abstracts, thus exposing the dynamic nature of the genre of RA abstracts in terms of their interpersonal dimension. The results show that the use of interactional metadiscourse has dropped in three consecutive decades, mainly as a result of a reduction in the number of boosters and attitude markers. How can these changes be explained? With regard to the use of boosters we may wonder why – if they are such an important means of expressing interpersonal relations in RA abstracts – they do become less frequent over the years. A plausible answer would be that the fall in the use of boosters and attitude markers is due to a converging move of (applied) linguistics towards the hard sciences. As Hyland (2005b, p. 187) shows, the use of interactional metadiscourse is markedly lower in the latter disciplines. It is not clear whether this move towards the hard sciences is a consequence of changing research practices, with a growing emphasis on empirical studies, or merely a change in rhetorical practices.

Alternatively, the changing combination of interactional features could be explained by a shift in rhetorical ethos. It may be argued that scholarly credibility is currently established by a deliberate, cautious expression of

[6] We owe this observation to one of the anonymous reviewers.

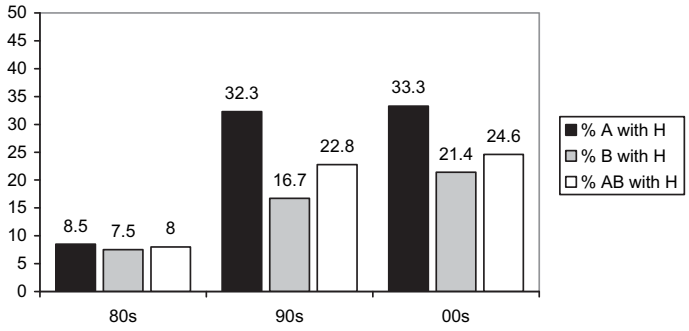


Fig. 5. Interaction between hedges and boosters/attitude markers.

scientific claims, rather than by the authoritative stance of an 'omniscient' academic. Obviously, these different types of ethos respectively correlate with a higher use of hedges on the one hand, and a higher use of attitude markers on the other hand. If the three decades of RA abstract writing that are investigated here are part of this more long-term, gradual shift in scholarly ethos, then the decrease in attitude markers and the strong position of hedges is in accordance with what we expect. Indeed, in the following two corpus examples from 1982, the authors' voice is quite outspoken. Expressions where they provide rather strong evaluative personal viewpoints are printed in bold:

(14) The information processing metaphor has gained a dominant status in psychological and linguistic studies of mind and language. However, there are a number of problems connected with this metaphor which **deserve** to be illuminated.
In this paper, I argue that the information processing paradigm forces us to postulate the existence of a **cryptic**, intermediate level between the phenomenological and the physical. This level is the result of a **confusion** of terms taken from the phenomenological and physical levels, as well as of the Cartesian dualistic picture of the human mind, which is at the basis of this confusion.
Furthermore, I suggest that the program analogy results from a **misunderstanding** of the way computers work. The development of a dispositional and phenomenological framework for the study of human language is **recommended** as an alternative to the information processing paradigm. (JoP 1982–8)

(15) (…) Given these concerns, I argue that grammars are quite **inadequate** for representing what we know about stories, and I claim that ''story grammar'' is at best an **overworked** and **misleading** metaphor. (JoP 1982–9)

In addition to the change in scholar ethos, the observation that hedges behave differently from boosters and attitude markers could be traced back to the growth of the discourse community. As the discourse community expands, the need for acknowledging stances other than the author's becomes more urgent. This can put a constraint on the use of boosters and attitude markers, two kinds of interactional elements that propagate the author's view.

There are two strategies to mitigate a strong presence of the author's stance: either by cutting down HBAs, or by taking the edge off and putting the boosters and attitude markers under the scope of an extra hedge. Both options seem to be exercised in our corpus. In Figure 3 a visible decline was shown in the overall use of attitude markers and boosters, and in Figure 4 it becomes clear that the attitude markers and boosters that are still used in recent RA abstracts are increasingly used with an accompanying hedge.

Needless to say, the results of this corpus study cannot be more than preliminary as all the abstracts come from a single journal. It remains to be seen whether the observed tendencies are corroborated by follow-up studies.

Of course, the quantitative study of the distribution of hedges, boosters and attitude markers does not cover the full range of what can be subsumed under interpersonality. The analysis of these three interactional categories is just one way of exploring the multifaceted nature of interpersonality in academic texts. Other angles, like the research on *evaluation* (see e.g. Hunston & Thompson, 2000; Mauranen & Bondi, 2003) may yield additional insights. Moreover,

the quantitative approach we have taken in the present study could be complemented by a qualitative inquiry into the way the interactional labels are applied (see Hyland, 2005a; Louhiala-Salminen, 1999; Markkanen, Steffensen, & Crismore, 1993). This could lead to further adaptations of Hyland's classification involving new categories or more subtle subcategories of metadiscourse. On the other hand, it could lead to a further integration of metadiscourse into other approaches dealing with interpersonality. Still, we hope to have shown that (a subset of) Hyland's classification of metadiscourse can serve as a working model to obtain empirical results about the way authors conceive their texts, in relation to their intended audience. Our results show that even in highly condensed genres like RA abstracts, interpersonal considerations are pervasive, and show striking diachronic dynamics.

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References

Ayers, G. (2008). The evolutionary nature of genre: an investigation of the short texts accompanying research articles in the scientific journal Nature. *English for Specific Purposes, 27*, 22–41.

Bazerman, C. (1984). Modern evolution of the experimental report in physics: spectroscopic articles in physical review 1893–1980. *Social Studies of Science, 14*(2), 163–196.

Biber, D. (2006). Stance in spoken and written university registers. *Journal of English for Academic Purposes, 5*, 97–116.

Dahl, T. (2004). Some characteristics of argumentative abstracts. *Akademisk Prosa, 2*, 49–69.

De Waard, A. (2009). Rhetorical Modeling of Scientific Research Papers. Paper presented at Rhetoric in Society. Second conference. University of Leiden, 21-23 January 2009.

Gillaerts, P., & Van de Velde, F. (2009). ''Het lijkt met name ook een goede strategie'': interactionele metadiscursiviteit in abstracts van onderzoeksartikelen. *Tijdschrift voor Taalbeheersing, 32*(2), 77–96.

Graetz, N. (1985). Teaching EFL students to extract structural information from abstracts. In J. Ulijn, & A. Pugh (Eds.), *Reading for professional purposes: Methods and materials in teaching languages* (pp. 123–135). Leuven: Acco.

Halliday, M. (1994). *An introduction to functional grammar* (2nd ed.). London: Arnold.

Hunston, S., & Thompson, G. (Eds.). (2000). *Evaluation in text: Authorial stance and the construction of discourse*. Oxford: Oxford University Press.

Hyland, K. (1998). Persuasion and context: the pragmatics of academic metadiscourse. *Journal of Pragmatics, 30*, 437–455.

Hyland, K. (2000). *Disciplinary discourses: Social interactions in academic writing*. London: Longman.

Hyland, K. (2005a). *Metadiscourse: Exploring writing in interaction*. London: Continuum.

Hyland, K. (2005b). Stance and engagement: a model of interaction in academic discourse. *Discourse Studies, 7*(2), 173–192.

Hyland, K., & Tse, P. (2004). Metadiscourse in academic writing: a reappraisal. *Applied Linguistics, 25*(2), 156–177.

Langacker, R. W. (1990). 'Subjectification'. *Cognitive Linguistics, 1*, 5–38.

Lorés, R. (2004). On RA abstracts: from rhetorical structure to thematic organisation. *English for Specific Purposes, 23*, 280–302.

Louhiala-Salminen, L. (1999). *From business correspondence to message exchange: The notion of genre in business communication*. Jyväskylä: University of Jyväskylä.

Markkanen, R., Steffensen, M. S., & Crismore, A. (1993). Quantitative contrastive study of metadiscourse. Problems in design and analysis of data. *Papers and Studies in Contrastive Linguistics, 23*, 137–151.

Mauranen, A., & Bondi, M. (2003). Evaluative language use in academic discourse. *Journal of English for Academic Purposes, 2*, 269–271.

Miller, C. R. (1984). Genre as social interaction. *Quarterly Journal of Speech, 70*, 151–167.

Salager-Meyer, F. (1990). Discourse flaws in medical English abstracts: a genre analysis per research- and text-type. *Text, 10*, 365–384.

Samraj, B. (2005). An exploration of a genre set: research article abstracts and introductions in two disciplines. *English for Specific Purposes, 24*, 141–156.

Stotesbury, H. (2003). Evaluation in research article abstracts in the narrative and hard sciences. *Journal of English for Academic Purposes, 2*, 327–341.

Swales, J. (1990). *Genre analysis: English in academic and research settings*. Cambridge: Cambridge University Press.

Thompson, G. (2001). Interaction in academic writing: learning to argue with the reader. *Applied Linguistics, 22*, 58–78.

Thompson, G., & Thetela, P. (1995). The sound of one hand clapping: the managment of interaction in written discourse. *Text, 15*, 103–127.

Ventola, E. (1994). Abstracts as an object of linguistic study. In S. Čmejrková, F. Daneš, & E. Havlová (Eds.), *Writing vs. speaking. Language, text, discourse, communication* (pp. 333–352). Tübingen: Gunter Narr.

Yakhontova, T. (2002). 'Selling' or 'telling'? The issue of cultural variation in research genres. In J. Flowerdew (Ed.), *Academic discourse* (pp. 216–232). Harlow: Pearson Education.

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